



STATISTICS

	ASSIGNMENT DETAILS
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SUBJECT	STATISTICS
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Mean :

The Arithmetic mean or Briefly the mean of the values (data) $x_1, x_2, \dots, x_n$ is defined as,

$$\begin{aligned}\text{mean}(\bar{x}) &= \frac{1}{n} (x_1 + x_2 + \dots + x_n) \\ &= \frac{1}{n} \sum_{i=1}^n x_i\end{aligned}\quad n \rightarrow \text{NO of Data}$$

If the Data have the frequency shown in the following table :

Variable (x_i)	x_1	x_2	...	x_n
Frequency (f_i)	f_1	f_2	...	f_n

Then the mean will be,

$$\begin{aligned}\bar{x} &= \frac{1}{N} (x_1 f_1 + x_2 f_2 + \dots + x_n f_n) \\ &= \frac{1}{N} \sum_{i=1}^n x_i f_i\end{aligned}$$

Theorem : If the two variables x and y are related by the equation,

$$y = \frac{x-c}{d}$$

then, $\bar{y} = \frac{\bar{x}-c}{d}$

c & d are any Numbers

Examples:

I) simple Distribution:

Q Find the mean of Data 5, 8, 3, 10, 12.

$$\begin{aligned} \text{Ans} \quad \text{Mean} &= \frac{1}{n} \sum x_i = \frac{1}{5} \times 38 \\ &= 7.6 \end{aligned}$$

II) ungrouped Distribution:

Q Find the mean of Data,

x_i	2	4	1	3	Total
f_i	3	2	4	1	10

$$\begin{aligned} \text{Ans} \quad \bar{x} &= \frac{1}{N} \sum x_i f_i \\ &= \frac{x_1 f_1 + x_2 f_2 + x_3 f_3 + x_4 f_4}{f_1 + f_2 + f_3 + f_4} \\ &= \frac{2 \times 3 + 4 \times 2 + 1 \times 4 + 3 \times 1}{10} \\ &= \frac{6 + 8 + 4 + 3}{10} = \frac{21}{10} = 2.1 \end{aligned}$$

III) Grouped frequency Distribution :

⊗ Find the Mean of following Data,

X:	110-119	120-129	130-139	140-149	150-159
f:	5	7	12	20	16
	160-169	170-179	180-189		
	10	7	3		

ANS →

Class interval	Modified class interval	Mid point (X _i)	Frequency (f _i)	$y_i = \frac{x_i - 154.5}{10}$ d = 10	f _i · y _i
110-119	109.5 - 119.5	115.5	5	-4	-20
120-129	119.5 - 129.5	124.5	7	-3	-21
130-139	129.5 - 139.5	134.5	12	-2	-24
140-149	139.5 - 149.5	144.5	20	-1	-20
150-159	149.5 - 159.5	C = 154.5	16	0	0
160-169	159.5 - 169.5	164.5	10	1	10
170-179	169.5 - 179.5	174.5	7	2	14
180-189	179.5 - 189.5	184.5	3	3	9
			$\Sigma f_i = 80$		$\Sigma y_i f_i = -52$

$$\bar{y} = \frac{\Sigma y_i f_i}{\Sigma f_i}$$

$$= \frac{-52}{80} = -0.65$$

$$y_i = \frac{x_i - c}{d}$$

$$\Rightarrow -0.65 = \frac{\bar{x} - 154.5}{10}$$

$$\Rightarrow \bar{x} = 154.5 - 6.5 = 148 \text{ (Ans)}$$

③ Find the Mean from the following Data :

Daily wages	25-29	30-34	35-39	40-44
No of workers	16	28	14	12

Modified Class Interval	Mid point	frequency (f _i)	y _i = $\frac{x_i - 37}{5}$	y _i f _i
24.5 - 29.5	27	16	-2	-32
29.5 - 34.5	32	28	-1	-28
34.5 - 39.5	37 = C	14	0	0
39.5 - 44.5	42	12	1	12
		$\Sigma f_i = 70$		$\Sigma y_i f_i = -48$

$$\bar{y} = \frac{\Sigma y_i f_i}{\Sigma f_i} = \frac{-48}{70} = \frac{-24}{35}$$

$$\bar{y} = \frac{\bar{x} - 37}{5}$$

$$\Rightarrow \bar{x} = 33.57$$

④ Calculate the Missing frequency from the following Table. Given, the mean is 7.74.

value	3-5	5-7	7-9	9-11	11-13	Total
frequency	32	-	57	-	25	200

Ans Let us consider that the missing frequency are f_1 & f_2 .

Class Interval	Mid point (x_i)	frequency (f_i)	$x_i f_i$
3-5	4	32	128
5-7	6	f_1	$6f_1$
7-9	8	57	456
9-11	10	f_2	$10f_2$
11-13	12	25	300
$\Sigma f_i = 114 + f_1 + f_2$			$\Sigma x_i f_i = 884 + 6f_1 + 10f_2$

Now,

$$114 + f_1 + f_2 = 200 \quad (\text{Given})$$

$$\Rightarrow f_1 + f_2 = 86 \quad \text{--- (I)}$$

Also,

$$\bar{x} = \frac{\Sigma x_i f_i}{\Sigma f_i}$$

$$\Rightarrow 7.74 = \frac{884 + 6f_1 + 10f_2}{200}$$

$$\Rightarrow 6f_1 + 10f_2 = 664$$

$$\Rightarrow 3f_1 + 5f_2 = 332 \quad \text{--- (II)}$$

from (I) and (II)

$$f_1 = 49$$

$$f_2 = 37 \quad (\text{ans})$$

Median

Median of a set of observations is the middle most value when the observations are arranged in increasing or decreasing order of magnitude. Thus to find the median of a set of observation is necessary to arrange the observation in order of magnitude.

There can be two cases while calculating median

i) If n is odd, then

$$\text{Median} = \left(\frac{n+1}{2} \right)$$

ii) If n is even, then

$$\text{Median} = \frac{1}{2} \left\{ \frac{n}{2} \text{th Observation} + \left(\frac{n}{2} + 1 \right) \text{th Observation} \right\}$$

I) Case I: Even $n \rightarrow$

* Find the Median of 5, 4, 7, 3, 21, 12

Ans Arranging in Ascending order,

3, 4, 5, 7, 12, 21

$$n = 6$$

$$\text{Median} = \frac{\frac{n}{2} + \left(\frac{n}{2} + 1 \right)}{2}$$

$$= \frac{\frac{6}{2} + \left(\frac{6}{2} + 1 \right)}{2}$$

$$= \frac{3^{\text{rd}} + 4^{\text{th}}}{2} = \frac{5 + 7}{2} = \frac{12}{2} = 6$$

ii) Case II : Odd $n \rightarrow$

⊗ Find the Median of 5, 8, 7, 20, 13, 3, 11.

Ans Arranging in Ascending Order

3, 5, 7, 8, 11, 13, 20

$$n = 7$$

$$\text{Median} = \frac{n+1}{2}$$

$$= \frac{7+1}{2} = 8/2 = 4^{\text{th}} \text{ Observation} = 8$$

iii) Case III : Ungrouped Frequency Distribution $\rightarrow$

⊗ Find the Median of following frequency Distribution,

$x :$	5	10	15	20	25	30	35	40
$f. :$	7	10	15	18	23	21	17	8

Ans

x_i	f_i	Cumulative frequency
5	7	7
10	10	$7+10 = 17$
15	15	$17+15 = 32$
20	18	$32+18 = 50$
25	23	$50+23 = 73$
30	21	$73+21 = 94$
35	17	$94+17 = 111$
40	8	$111+8 = 119$

$$N = 119$$

$$\frac{N+1}{2} = \frac{119+1}{2} = \frac{120}{2} = 60$$

$$73 > 60$$

So Median is 25.

(Now we have to check the cumulative frequency and find the Number > 60 . Then the corresponding x_i will be the Median)

IV) Case IV : Grouped frequency Distribution $\rightarrow$

④ Calculate Median from the following frequency Distribution.

Height	60-63	64-67	68-71	72-75	76-79	80-83
No. of students	8	3	18	6	16	8

Ans.

Class Interval	Modified class Interval	frequency (f_i)	cumulative frequency
60-63	59.5-63.5	8	8
64-67	63.5-67.5	3	11
68-71	67.5-71.5	18	29
72-75	71.5-75.5	6	(35)
76-79	75.5-79.5	16	51
80-83	79.5-83.5	8	59

$$N = 59$$

$$\text{Median} = L_m + \frac{\frac{N}{2} - F}{f_m} \times i$$

where,

$L_m \rightarrow$ Lower Boundary of Median class

$N \rightarrow$ Total frequency

$F \rightarrow$ cumulative frequency of the class preceding to the median class

$f_m \rightarrow$ frequency of median class

$i \rightarrow$ width of median class.

$$\frac{N}{2} = \frac{59}{2} = 29.5$$

Just greater than 29.5 is, 35

So the median class is, (71.5-75.5)

$$l_m \rightarrow 71.5$$

$$F \rightarrow 29$$

$$f_m \rightarrow 6$$

$$i = (75.5 - 71.5) = 4$$

$$N/2 \rightarrow 29.5$$

$$\begin{aligned} \text{Median} &\rightarrow \left(71.5 + \frac{29.5 - 29}{6} \times 4 \right) \\ &= 71.83 \end{aligned}$$

Mode

The Observation having maximum frequency is called Mode.

I) Case I : ungrouped frequency Distribution $\rightarrow$
① Calculate the mode for following Data,
1, 12, 5, 8, 12, 13, 8, 1, 4, 8, 7, 8, 5.

Sol

x	f
1	2
4	1
5	2
7	1
8	4
12	2
13	1

The Mode of this distribution is $\rightarrow 8$.

II) Case II : Grouped frequency Distribution $\rightarrow$
 * Calculate mode from the following frequency Distribution :

Height	10-15	15-20	20-25	25-30	30-35
No of students	6	9	11	7	7

Ans Here the greatest frequency lies in class 20-25.

Hence the Modal Class is 20-25

$$\text{Mode} = l_m + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i$$

$$l_m \rightarrow 20$$

$$f_1 \rightarrow 11$$

$$f_0 \rightarrow 9$$

$$f_2 \rightarrow 7$$

$$i = (25-20) \\ = 5$$

$l_m \rightarrow$ lowest Boundary of Modal class

$f_1 \rightarrow$ frequency of Modal class

$f_0 \rightarrow$ frequency of the class preceding the modal class

$f_2 \rightarrow$ frequency of the class succeeding the modal class

$i \rightarrow$ width of modal class

So, the Mode is

$$= 20 + \frac{11 - 9}{2 \times 11 - 9 - 7} \times 5 \\ = 21.67$$

▣ NOTE: Relation Between Mean, Median and Mode (for single mode only) $\rightarrow$

$$\text{Mean} - \text{Mode} \cong 3 (\text{Mean} - \text{Median})$$

Variance & Standard Deviation

Variance: The Mean of the squares of the Differences on the observations (or Data) assumed by a variable from of their arithmetic mean is called variance of the variable.

Standard Deviation: The positive square root of Variance is called standard Deviation (S.D). Thus if the Data is, $x_1, x_2 \dots x_n$, then Variance is,

$$\begin{aligned} \text{Var}(x) &= \frac{1}{n} \{ (x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2 \} \\ &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 \end{aligned}$$

$$\text{Standard Deviation } \sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

I) Case I : Simple Distribution $\rightarrow$

① Find the variance of the variable assuming the Data 20, 85, 120, 60, 40.

Ans

$$\text{Mean } (\bar{x}) = \frac{20+85+120+60+40}{5} \\ = 65$$

x_i	$x_i - \bar{x} = x_i - 65$	$(x_i - 65)^2$
20	$20 - 65 = -45$	2025
85	$85 - 65 = 20$	400
120	55	3025
60	-5	25
40	-25	625

$$\Sigma = 6100$$

$$\text{Var}(x) = \frac{1}{n} \Sigma (x_i - \bar{x})^2$$

$$\Rightarrow \text{Var}(x) = \frac{\Sigma (x_i - 65)^2}{n}$$

$$= \frac{6100}{n} = \frac{6100}{5} = 1220$$

$$\text{S.D} = \sigma_x = +\sqrt{\text{Var}(x)}$$

$$= +\sqrt{1220}$$

$$= 34.93$$

II) Case II: Ungrouped Simple frequency Distribution $\rightarrow$

Variable (x_i)	94.5	84.5	74.5	64.5	54.5	44.5
Frequency (f_i)	2	12	22	20	14	4
	34.5					
	1					

Now, find the S.D & Variance of the Data.

Ans

x_i	f_i	$y_i = \frac{x_i - 64.5}{10}$	y_i^2	$f_i y_i$	$f_i y_i^2$
94.5	2	3	9	6	18
84.5	12	2	4	24	48
74.5	22	1	1	22	22
64.5	20	0	0	0	0
54.5	14	-1	1	-14	14
44.5	4	-2	4	-8	16
34.5	1	-3	9	-3	9

$$N = 75$$

$$\sum f_i y_i = 27$$

$$\sum f_i y_i^2 = 127$$

$$\begin{aligned} \text{Var}(y) &= \frac{1}{N} \sum f_i y_i^2 - \left(\frac{1}{N} \sum f_i y_i \right)^2 \\ &= \frac{127}{75} - \left(\frac{27}{75} \right)^2 \\ &= 1.563733 \end{aligned}$$

$$\sigma_y = +\sqrt{\text{Var}(y)} = \sqrt{1.563733} = 1.2519318$$

$$\therefore \sigma_y = \frac{\sigma_x}{10}$$

$$\Rightarrow 1.2519 \times 10$$

$$\Rightarrow 12.5$$

$$\begin{aligned} \text{Var}(x) &= (\sigma_x)^2 \\ &= (12.5)^2 \\ &= 156.25 \end{aligned}$$

III) Case III : Grouped frequency Distribution →
 * Find the Variance and Standard Deviation of the following frequency Distribution

Weight	36-40	41-45	46-50	51-55	56-60	61-65
No of Persons	14	26	40	33	50	37
	66-70					
	25					

Ans

Class Interval	Modified class Interval	Mid point (x_i)	frequency (f_i)	$y = \frac{x_i - 53}{5}$	y_i^2
36-40	35.5-40.5	38	14	-3	9
41-45	40.5-45.5	43	26	-2	4
46-50	45.5-50.5	48	40	-1	1
51-55	50.5-55.5	53	33	0	0
56-60	55.5-60.5	58	50	1	1
61-65	60.5-65.5	63	37	2	4
66-70	65.5-70.5	68	25	3	9

$N = 225$

$f_i y_i$	$f_i y_i^2$
-42	126
-52	104
-40	40
0	0
50	50
74	148
75	225
65	693

Now,

$$\text{var}(y) = \frac{1}{N} \sum f_i y_i^2 - \left(\frac{1}{N} \sum f_i y_i \right)^2$$

$$= \frac{693}{225} - \left(\frac{65}{225} \right)^2$$

$$= 2.996$$

$$\sigma_y = \sqrt{2.996} = 1.731$$

$$\sigma_y = \frac{\sigma_x}{5}$$

$$\text{var}(x) = (\sigma_x)^2$$

$$= 8.655^2$$

$$= 74.909025$$

$$\Rightarrow \sigma_x = 8.655$$

Coefficient of Variation

The Measure of Central Tendency represents the set of all Observations/Data of a variable and Standard Deviation shows the Measure of Consistency of the Data. In fact to compare the consistency of data of two variables we have to find the Coefficient of Variation (C.V) of two Data. The variable having greater C.V is lesser consistent.

$$C.V = \frac{\text{Standard Deviation}}{\text{Mean}} \times 100$$

④ The scores of two batsman A and B in 10 innings are:

A	19	31	48	53	67	90	10	62	40	80
B	32	28	47	63	71	39	10	60	96	14

Find which Batsman is more consistent in Scoring.

Ans

A		
x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
19	$(19 - 50) = -31$	961
31	$(31 - 50) = -19$	361
48	$(48 - 50) = -2$	4
53	$\vdots$	$\vdots$
67	$\vdots$	$\vdots$
90	$\vdots$	$\vdots$
10	$\vdots$	$\vdots$
62	$\vdots$	$\vdots$
40	$\vdots$	$\vdots$
80	$\vdots$	$\vdots$
500		5953.6

$$\begin{aligned}\bar{x} &= \frac{\sum x_i}{n} \\ &= \frac{500}{10} \\ &= 50\end{aligned}$$

B		
y_i	$y_i - \bar{y}$	$(y_i - \bar{y})^2$
32	$(32 - 46) = -14$	196
28	$(28 - 46) = -18$	324
47	$(47 - 46) = 1$	1
63	$\vdots$	$\vdots$
71	$\vdots$	$\vdots$
39	$\vdots$	$\vdots$
10	$\vdots$	$\vdots$
60	$\vdots$	$\vdots$
96	$\vdots$	$\vdots$
14	$\vdots$	$\vdots$
460		6502.5

$$\begin{aligned}\bar{y} &= \frac{\sum y_i}{n} \\ &= \frac{460}{10} \\ &= 46\end{aligned}$$

So,

$$\begin{aligned}\text{C.V of } x &= \frac{\sigma_x}{\bar{x}} \times 100 \\ &= \frac{24.4}{50} \times 100 \\ &= 48.8\end{aligned}$$

$$\begin{aligned}\sigma_x &= \sqrt{\frac{1}{n} \sum (x_i - \bar{x})^2} \\ &= \sqrt{\frac{5953.6}{10}} \\ &= 24.4\end{aligned}$$

$$\begin{aligned}\text{C.V of } y &= \frac{\sigma_y}{\bar{y}} \times 100 \\ &= \frac{25.5}{46} \times 100 \\ &= 55.43\end{aligned}$$

$$\begin{aligned}\sigma_y &= \sqrt{\frac{1}{n} \sum (y_i - \bar{y})^2} \\ &= \sqrt{\frac{6502.5}{10}} \\ &= 25.5\end{aligned}$$

As we know less C.V means more consistency.
So from the result we can say that batsman A is more consistent.

Moments

The k^{th} moment of the values (data) $x_1, x_2, \dots, x_n$ about a number a is defined as,

$$\begin{aligned}&\frac{1}{n} \{ (x_1 - a)^k + (x_2 - a)^k + \dots + (x_n - a)^k \} \\ &= \frac{1}{n} \sum_{i=1}^n (x_i - a)^k\end{aligned}$$

If the Data have the frequency shown in the following table :

x_i	x_1	x_2	x_3	...	x_n
f_i	f_1	f_2	f_3	...	f_n

$$\frac{1}{N} (f_1 (x_1 - a)^k + f_2 (x_2 - a)^k + \dots + f_n (x_n - a)^k)$$

$$= \frac{1}{N} \sum_{i=1}^n f_i (x_i - a)^k$$

These are two types of moments

I) Central Moment :

Any moment about the A.M of the Data is called central moment. The k^{th} Order central Moment is called m_k or μ_k .

II) Raw Moment :

Any moment about 0 is called raw moment, and this is Denoted by m'_k .

■ Relation between Central Moment and any moment :

If m_k k^{th} Central moment and m'_k is k^{th} moment about any number then,

$$I) m_2 = m'_2 - m_1'^2$$

$$II) m_3 = m'_3 - 3m'_2 m'_1 + 2m_1'^3$$

$$III) m_4 = m'_4 - 4m'_3 m'_1 + 6m'_2 m_1'^2 - 3m_1'^4$$

If two variables x and y are Related as,
 $y = \frac{x - c}{d}$ where c, d are constants. then

$$m_{k1}(y) = \frac{m_k(x)}{d^k}$$

④ The first four moments of a distribution about 5 are 2, 10, 40 and 218. Find the first four central moments about 0.

Solⁿ Let, m_1 and m_1' be central moments about '5' respectively.

using the relation between m_1 and m_1' we have,

$$m_2 = 10 - 2^2 = 6 \quad m_3 = 40 - 3 \times 10 \times 2 + 2 \times 2^3 = -4$$

$$m_4 = 90$$

Let $\mu_1, \mu_2, \mu_3, \mu_4$ be the moments about '0'

Now, 1st moment about '5' = 2

$$\Rightarrow \frac{1}{n} \sum (x_i - 5) = 2$$

$$\Rightarrow \frac{1}{n} \sum x_i - \frac{1}{n} \sum 5 = 2$$

$$\Rightarrow \frac{1}{n} \sum x_i - \frac{1}{n} \times 5n = 2$$

$$\Rightarrow \frac{1}{n} \sum x_i = 7$$

$$\Rightarrow \mu_1 = 7$$

2nd order moment about '5' = 10

$$\Rightarrow \frac{1}{n} \sum (x_i - 5)^2 = 10$$

$$\Rightarrow m_2 = \mu_2 - \mu_1^2$$

$$\Rightarrow 6 = \mu_2 - 7^2$$

$$\Rightarrow \mu_2 = 55$$

3rd Order moment about '5' = 40

$$m_3 = \mu_3 - 3\mu_2\mu_1 + 2\mu_1^3$$

$$\Rightarrow -4 = \mu_3 - 3\mu_2\mu_1 + 2\mu_1^3$$

$$\Rightarrow \mu_3 = 465$$

similarly,

$$\mu_4 = 4143$$

Skewness

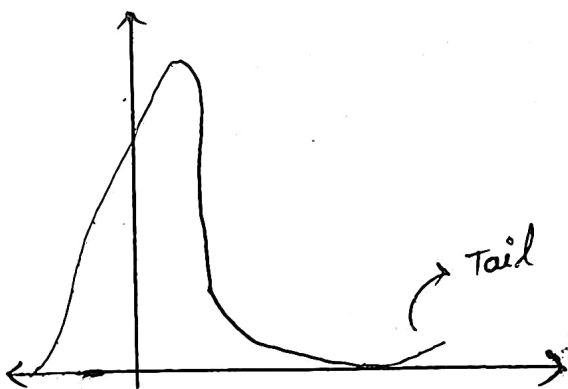
The Skewness of a frequency Distribution
are,

$$\text{Co-efficient of skewness } (\gamma_1) = \frac{m_3}{\sigma^3}$$

(amount of asymmetry)

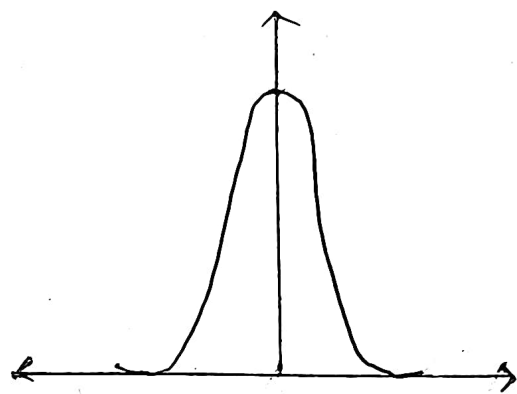
Significance of Skewness

Case I:



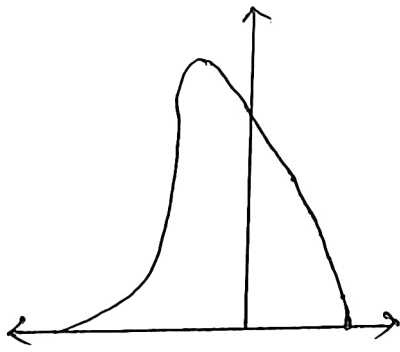
coefficient of skewness > 0
Positively skewed

Case II:



coefficient of skewness
= 0

Case III :



co-efficient of skewness < 0
negatively skewed

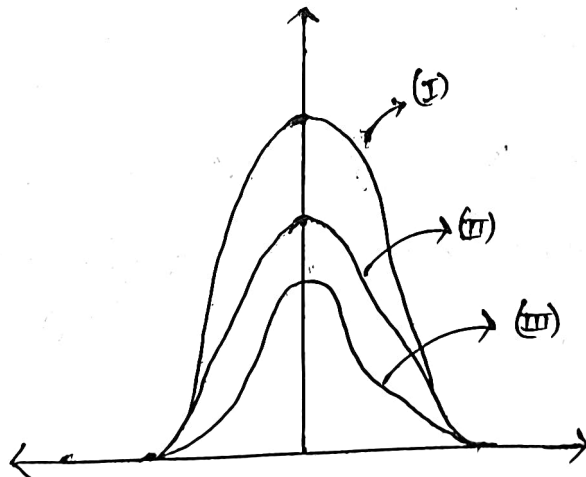
Kurtosis

Kurtosis tells the position of Graph in Y Axis. The kurtosis of a frequency Distribution are,

$$\text{co-efficient of kurtosis } (\gamma_2) = \frac{m_4}{\sigma_4}$$

$$m_4 = \frac{1}{N} \sum (x_i - \bar{x})^4 f_i$$

There can be Three cases of kurtosis,



Case (I) : If Peakness > 3 , Then it's Called Leptokurtic Distribution

Case (II) : If Peakness $= 3$, Then it's Called Mesokurtic Distribution

Case (III) : If Peakness < 3 , Then it's called platikurtic Distribution.

* Calculate Skewness & Kurtosis of following Distribution :

Height	60-63	64-67	68-71	72-75	76-79	80-83
Students	8	3	18	6	16	8

Class Interval	Modified Interval	Mid Point	frequency (f_i)	$y = \frac{x_i - c}{d}$ $d=4$	$f_i y_i$	$f_i y_i^2$	$f_i y_i^3$	$f_i y_i^4$
60-63	59.5-63.5	61.5	8	-3	-24	72	-216	648
64-67	63.5-67.5	65.5	3	-2	-6	12	-24	48
68-71	67.5-71.5	69.5	18	-1	-18	18	-18	18
72-75	-	(73.5)	6	0	0	0	0	0
76-79	-	77.5	16	1	16	16	16	16
80-83	-	81.5	8	2	16	32	64	124
				59	-16	150	-178	794

Now, Raw Moments,

$$m'_1(y) = \frac{1}{N} \sum f_i y_i = \frac{-16}{59}$$

$$m'_2(y) = \frac{\sum f_i y_i^2}{N} = \frac{150}{59}$$

$$m'_3(y) = \frac{\sum f_i y_i^3}{N} = \frac{-178}{59}$$

$$m'_4(y) = \frac{\sum f_i y_i^4}{N} = \frac{794}{59}$$

Central Moments,

$$m_1(y) = 0 \longrightarrow \text{Always } 0$$

$$\begin{aligned} m_2(y) &= m_2'(y) - (m_1'(y))^2 \\ &= \frac{150}{59} - \left(\frac{-16}{59}\right)^2 \\ &= 2.468830 \end{aligned}$$

$$\begin{aligned} m_3(y) &= m_3'(y) - 3m_2'(y)m_1'(y) + 2\{m_1'(y)\}^3 \\ &= -\frac{178}{59} - 3 \times \frac{150}{59} \times \left(\frac{-16}{59}\right) + 2 \times \left(\frac{-16}{59}\right)^3 \\ &= -0.988465 \end{aligned}$$

$$\begin{aligned} m_4(y) &= m_4'(y) - 4m_3'(y) + 6m_2'(y)(m_1'(y))^2 \\ &\quad - 3(m_1'(y))^4 \\ &= \frac{794}{59} - 4 \times \left(\frac{-178}{59}\right) \times \left(\frac{-16}{59}\right) + 6 \times \frac{150}{59} \times \left(\frac{-16}{59}\right)^2 \\ &\quad - 3 \times \left(\frac{-16}{59}\right)^4 \\ &= 11.290607 \end{aligned}$$

Moment with respect to x ,

$$\begin{aligned} m_1(x) &= 0 & m_2(y) &= \frac{m_2(x)}{d^2} \\ & & \Rightarrow 2.468830 &= \frac{m_2(x)}{42} \\ & & \Rightarrow m_2(x) &= 39.501 \end{aligned}$$

$$\begin{aligned} m_3(y) &= \frac{m_3(x)}{d^3} & m_4(y) &= \frac{m_4(x)}{d^4} \\ \Rightarrow -63.26177 & & \Rightarrow 2890.3453 \end{aligned}$$

$$m_2(x) = \text{Var } x = 39.50128$$

$$\Rightarrow \sigma = \sqrt{39.50128}$$

$$\text{So, Skewness of } x = \frac{m_3(x)}{(\sigma_x)^3} = \frac{-63.26177}{(\sqrt{39.5012})^3} = -0.2548$$

$$\text{Kurtosis of } x = \frac{m_4(x)}{(\sigma_x)^4} = \frac{2890.39592}{(\sqrt{39.5012})^4} = 11.6424$$

Correlation

Bivariate Data : Let x, y be two random variables defined on an event space S of an experiment. Then the pair

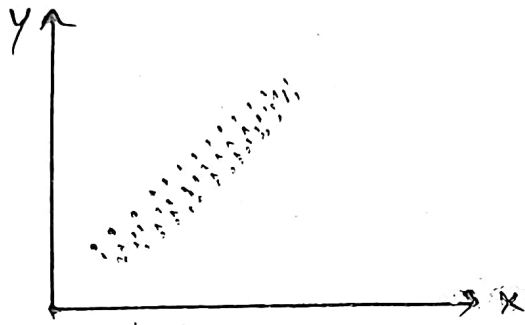
(x, y) is called a Bivariate on the same event space. Such bi-variate (x, y) assumes a pair of results (x, y) corresponding to an outcome in the event space where x and y are assumed by the random variables x and y respectively against the same outcome. These pair of values (x, y) are called Bi-Variate Data.

□ Different type of Bivariate Data :

- I) Both Discrete
- II) one is Continuous and one is Discrete
- III) Both are Continuous.

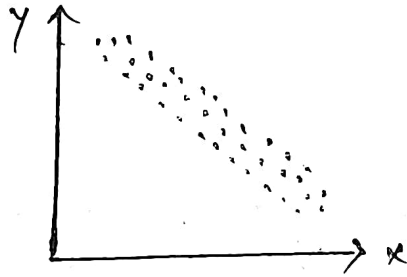
Different type of Correlation :

I) Positively Correlated : If y has a tendency to increase as x increases we say x and y are positively correlated.



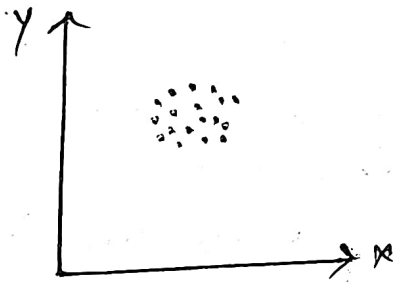
$$r/f \geq 0$$

II) Negatively Correlated : If y has a tendency to decrease as x increases we say x and y are negatively correlated.



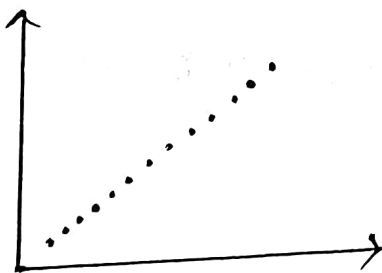
$$r/f \leq 0$$

III) Uncorrelated or Zero Correlated : If the value of y are not affected by the changes in the values of x , then it's called zero correlated.



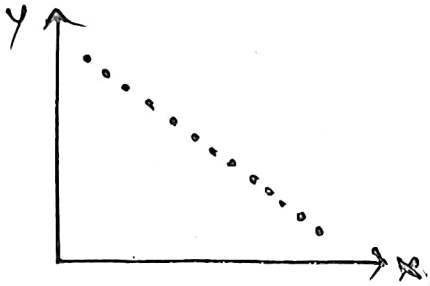
$$r/f = 0$$

IV) Strictly Positively Correlated : If y strictly increases as x increases, then it's called x and y are strictly positively correlated.



$$r/f > 0$$

iv) Strictly Negatively Correlated: If y strictly decrease as x increases we say x and y are strictly negatively correlated.



$$r/f < 0$$

Covariance & Correlation: Let $(x_1, y_1), (x_2, y_2) \dots (x_n, y_n)$ be a bivariate data set where $n \geq 2$. Then

The Covariance between the variables x and y is defined as,

$$\begin{aligned} \text{Cov}(x, y) &= \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \\ &= \frac{1}{n} \sum_{i=1}^n x_i y_i - \bar{x} \bar{y} \end{aligned}$$

The Correlation between them is defined as

$$r_{xy} = \rho_{xy} = \text{Corr}(x, y)$$

$$= \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x) \text{Var}(y)}} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y}$$

This definition is Given by Pearson and so this is also known as Pearson's product Moment Correlation Coefficient.

Properties of Covariance:

Let $(x_1, y_1), (x_2, y_2), \dots, (x_m, y_m)$ be a bivariate data set. Let $a, b, c, d \in \mathbb{R}$ be any numbers, where $a, c \neq 0$. Then,

- ① $\text{Cov}(ax+b, cy+d) = ac \text{Cov}(x, y)$
- ② $\text{Cov}(x, y) = \text{Cov}(y, x)$
- ③ $\text{Cov}(x, x) = \text{Var}(x)$
- ④ $\text{Cov}(ax+by, cx+dy) = ac \text{Var}(x) + (ad+bc) \text{Cov}(x, y) + bd \text{Var}(y)$
- ⑤ $\text{Var}(x+y) = \text{Var}(x) + 2\text{Cov}(x, y) + \text{Var}(y)$, $\text{Cov}(x+y, x-y) = \text{Var}(x) - \text{Var}(y)$

Properties of Correlation Coefficient:

- ① $r_{xy} = r_{yx}$
- ② $\text{Cov}(x, y)$ & σ_x, σ_y have same unit.
- ③ Greater the values of r_{xy} , strength of association between x and y is greater.
- ④ Let (x, y) and (u, v) represent two sets of bivariate data such that, $u = ax+b$ and $v = cy+d$ then,

$$r_{uv} = \frac{ac}{|a| |c|} r_{xy}$$

⑤ Let (x, y) and (u, v) represent two sets of bivariate data, such that $u = ax + b$ & $v = cy + d$. If a, c have same sign then, $r_{uv} = r_{xy}$, If a, c have opposite sign then $r_{uv} = -r_{xy}$

⑥
$$\text{Var}(x+y) = \text{Var}(x) + 2\sigma_x \sigma_y r_{xy} + \text{Var}(y)$$

$$= \text{Var}(x) - 2\sigma_x \sigma_y r_{xy} + \text{Var}(y)$$

⑦ $-1 \leq r_{xy} \leq 1$

⑧ If $r_{xy} = \pm 1$ the max and the min value of Correlation Coefficient, then y is a linear function of x and vice versa.

* Find the Correlation Coefficient between two variables X and Y where the bivariate data are Given by :

x	65	66	67	67	68	69	70	72
y	67	68	65	68	72	72	69	71

Ans

x_i	y_i	$u_i = x_i - 67$	$v_i = y_i - 68$	u_i^2	v_i^2	$u_i v_i$
65	67	-2	-1	4	1	2
66	68	-1	0	1	0	0
67	65	0	-3	0	9	0
67	68	0	0	0	0	0
68	72	1	4	1	16	4
69	72	2	4	4	16	8
70	69	3	1	9	1	3
72	71	5	3	25	9	15
		$\Sigma 8$	$\Sigma 8$	$\Sigma 44$	$\Sigma 52$	$\Sigma 32$

Here,
 $n = 8$

$$\begin{aligned}\text{Cov}(u, v) &= \frac{1}{n} \sum u_i v_i - \bar{u} \bar{v} \\ &= \frac{1}{n} \sum u_i v_i - \left(\frac{1}{n} \sum u_i \right) \left(\frac{1}{n} \sum v_i \right) \\ &= \frac{1}{8} \times 32 - \left(\frac{1}{8} \times 8 \right) \left(\frac{1}{8} \times 8 \right) \\ &= 4 - 1 = 3\end{aligned}$$

$$\begin{aligned}\text{Var}(u) &= \sigma_u^2 = \frac{1}{n} \sum u_i^2 - \left(\frac{1}{n} \sum u_i \right)^2 \\ &= \frac{1}{8} \times 44 - \left(\frac{1}{8} \times 8 \right)^2 \\ &= 4.5\end{aligned}$$

$$\begin{aligned}\text{Var}(v) &= \sigma_v^2 = \frac{1}{n} \sum v_i^2 - \left(\frac{1}{n} \sum v_i \right)^2 \\ &= 5.5\end{aligned}$$

$$\begin{aligned}\text{Now, } r_{uv} &= \frac{\text{Cov}(u, v)}{\sigma_u \sigma_v} \\ &= \frac{3}{\sqrt{4.5} \times \sqrt{5.5}} = 0.603\end{aligned}$$

$$\begin{aligned}u &= x - 67 & v &= y - 68 \\ u &= ax + b & v &= cy + d\end{aligned}$$

□ If value of a and c both are of same sign, the sign of r_{xy} will be positive, else it will be negative

$$\text{Here, } a = 1 \quad c = 1,$$

$$\text{So, } r_{xy} = 0.603$$

④ If $u+3x=5$ & $2y-v=7$ and correlation coefficient is 0.12 (of x and y). Then find the correlation coefficient of u and v .

Ans

Here,

$$u+3x=5$$

$$\text{or, } u = -3x+5$$

$$u = ax+b$$

$$a = -3$$

$$2y-v=7$$

$$\Rightarrow v = 2y-7$$

$$v = cy+d$$

$$c = +2$$

Now, we know that, if sign of a & c are different, the sign of σ_{uv} becomes negative.

$$\text{So, } \sigma_{uv} = -0.12$$

⑤ If x and y are two variables, determine the value of k , such that, $(x+ky)$ and $(x+\frac{\sigma_x}{\sigma_y}y)$ are uncorrelated.

Ans

Here,

$$u = ax+by$$

$$v = cx+dy$$

$$\text{Cov}(u,v) = ac \text{Var}(x) + (ad+bc) \text{Cov}(x,y) + db \text{Var}(y)$$

Let,

$$u = x+ky \quad \& \quad v = x + \frac{\sigma_x}{\sigma_y} y$$

Since $\sigma_{uv} = 0$ (Given, they are uncorrelated)

$$\text{Then } \Rightarrow \text{Cov}(u,v) = 0$$

$$\left[\begin{array}{l} \sigma_{uv} = \frac{\text{Cov}(u,v)}{\sigma_u \sigma_v} \\ \Rightarrow \text{Cov}(u,v) = 0 \end{array} \right]$$

$$a = 1, b = k, c = 1, d = \frac{\sigma_x}{\sigma_y}$$

$$\therefore \text{Cov}(u, v) = 1 \times 1 \sigma_x^2 + \left(1 \cdot \frac{\sigma_x}{\sigma_y} + k \cdot 1\right) \text{Cov}(x, y) + k \cdot \frac{\sigma_x}{\sigma_y} \cdot \sigma_y^2$$

$$\Rightarrow 0 = \sigma_x^2 + \left(\frac{\sigma_x}{\sigma_y} + k\right) \sigma_{xy} \sigma_x \sigma_y + k \sigma_x \sigma_y$$

$$\Rightarrow 0 = \sigma_x^2 + \left(\frac{\sigma_x + k \sigma_y}{\sigma_y}\right) \sigma_{xy} \sigma_x \sigma_y + k \sigma_x \sigma_y$$

$$\left[\begin{array}{l} \sigma_{xy} = \frac{\text{Cov}(x, y)}{\sigma_x \sigma_y} \\ \Rightarrow \text{Cov}(x, y) = \sigma_{xy} \sigma_x \sigma_y \end{array} \right]$$

$$\text{or, } 0 = \sigma_x^2 + \sigma_{xy} \sigma_x^2 + k \sigma_{xy} \sigma_x \sigma_y + k \sigma_x \sigma_y$$

$$\Rightarrow k \sigma_x \sigma_y (\sigma_{xy} + 1) = -\sigma_x^2 - \sigma_{xy} \sigma_x^2$$

$$\Rightarrow k = \frac{-\sigma_x^2 (1 + \sigma_{xy})}{\sigma_x \sigma_y (1 + \sigma_{xy})}$$

$$\Rightarrow k = -\frac{\sigma_x}{\sigma_y}$$

* If $\text{Var}(x) = 9$, $\text{Var}(y) = 4$ and $\text{Var}(x - y) = \text{Var}(x)$, find the Correlation Coefficient between x and y .

$$\text{Var}(x - y) = \text{Var}(x) + \text{Var}(y) - 2 \text{Cov}(x, y)$$

$$\text{or, } \text{Var}(x) = \text{Var}(x) + 4 - 2k \sigma_{xy}$$

$$\Rightarrow \sigma_{xy} = \frac{4}{12} = \frac{1}{3} \text{ (Ans)}$$

$$\left[\begin{array}{l} \sigma_{xy} = \frac{\text{Cov}(x, y)}{\sqrt{\text{Var}(x) \text{Var}(y)}} \\ \Rightarrow \frac{\text{Cov}(x, y)}{\sqrt{4 \times 9}} = \sigma_{xy} \\ \Rightarrow \text{Cov}(x, y) = 6 \sigma_{xy} \end{array} \right]$$

Rank Correlation

Sometimes for a bivariate data (x, y) x and y can't be measured quantitatively but qualitative assessment can be done by allotting rank to the individuals. The Correlation Coefficient of the two series of Rank is called Rank Correlation Coefficient of x and y .

If two series of Rank are Given in the characteristics x and y then Rank Correlation Coefficient = $\left(1 - \frac{6 \sum d^2}{n^3 - n}\right)$

- ⑩ In a beauty Competition two Judges I and II rank the entries as follows. Find the rank Correlation between the Judgement of the two Judge.

$x :$	1	2	3	4	5	6	7	8	9	10	11	12
$y :$	12	9	6	10	3	5	4	7	8	2	11	1

Ans

R_x	R_y	$d_r = R_x - R_y $	d^2
1	12	11	121
2	9	7	49
3	6	3	9
4	10	6	36
5	3	2	4
6	5	1	1
7	4	3	9
8	7	1	1
9	8	1	1
10	2	8	64
11	11	0	0
12	1	11	121

$$\sum d^2 = 416$$

Now,

Rank Correlation Coefficient,

$$= \left(1 - \frac{6 \times 416}{12^3 - 12} \right) \quad n = 12$$

$$= -0.4545$$

Since Rank Correlation Coefficient is Negative, we may conclude that if the Rank given by Judge 1 is higher, then there is a tendency that Judge 2 will give lower Rank.

- ⑩ In a Class, ten students obtained the following marks in mathematics and Physics out of 100 each :

Math	8	36	98	25	75	82	92	62	65	35
Physics	84	51	91	60	68	62	86	58	35	49

Find the Rank Correlation Coefficient of the marks, obtained in the two subjects.

Ans

X (Math)	Y (Physics)	R_x	R_y	$d = R_x - R_y $	d^2
8	84	10	3	7	49
36	51	7	8	1	1
98	91	1	1	0	0
25	60	9	6	3	9
75	68	4	4	0	0
82	62	3	5	2	4
92	86	2	2	0	0
62	58	6	7	1	1
65	35	5	10	5	25
35	49	8	9	1	1

$$\sum d^2 = 90$$

Now, the Rank Correlation Coefficient

$$= \left(1 - \frac{6 \times 90}{10^3 - 10} \right)$$

$$= 0.455$$

⑧ Marks Obtained by 10 students in physics and math are given in the following table,

physics	48	33	40	9	16	16	65	24	16	57
Math	13	13	24	6	15	4	20	9	6	19

Find the Rank Correlation Coefficient of the two series of Marks.

Ans

X	Y	R_x	R_y	$d = R_x - R_y $	d^2
48	13	3	5.5	2.5	6.25
33	13	5	5.5	0.5	0.25
40	24	4	1	3	9
9	6	10	8.5	1.5	2.25
16	15	8	4	4	16
16	4	8	10	2	4
65	20	1	2	1	1
24	9	6	7	1	1
16	6	8	8.5	0.5	0.25
57	19	2	3	1	1

$$\sum d^2 = 41$$

$$\frac{5+6}{2} = 5.5$$

$$\frac{7+8+9}{3} = 8$$

$$\frac{8+9}{2} = 8.5$$

$$F = F_{16} + F_{13} + F_6$$

$$= \frac{3^3 - 3}{12} + \frac{2^3 - 2}{2} + \frac{2^3 - 2}{12}$$

$$= 3$$

$$F = \sum \frac{t^3 - t}{12}$$

t = No of Occurance

Now,

$$\begin{aligned}\text{Rank Correlation} &= 1 - \frac{6(\sum d^2 + F)}{n^3 - n} \\ &= 1 - \frac{6 \times (41 + 3)}{10^3 - 10} \\ &= 0.733\end{aligned}$$

Regression

Regression: From the bivariate data Given by (x, y) the estimation or prediction; of the average value of a variable, say y , Corresponding to a specified value of the other variable x is called regression. The Average value of x may also be predicted Corresponding to a specified value of y .

Regression Line: From the bivariate Data Given by (x, y) the regression i.e. the prediction of average value of one variable against other is done by a suitable curve or straight line which fits best the scatter diagram of the Data. This line is called Regression line or least square regression line.

Equation of Regression Line:

Let $(x_1, y_1), (x_2, y_2), (x_3, y_3), \dots, (x_n, y_n)$ be n pairs of observations Given by the bivariate data (x, y) . Let $(\bar{x}, \bar{y})$ be the means.

σ_x, σ_y be the standard deviations of the values of x and y respectively; r_{xy} is Correlation Coefficient. Then the equation of the

I) Regression Line of y on x is,

$$(y - \bar{y}) = b_{yx} (x - \bar{x}), \text{ where, } b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x}$$

II) Regression Line of x on y is,

$$(x - \bar{x}) = b_{xy} (y - \bar{y}), \text{ where, } b_{xy} = r_{xy} \frac{\sigma_x}{\sigma_y}$$

Regression Coefficient

I) The quantity $b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x}$ appeared in the 'regression equation of y on x ' is called regression coefficient of y on x .

II) The quantity $b_{xy} = r_{xy} \frac{\sigma_x}{\sigma_y}$ appeared in the 'regression equation of x on y ' is called regression coefficient of x on y .

Properties of Regression Line & Coefficient

I) for a Bivariate Data Given by (x, y) the Correlation Coefficient and the Regression Coefficients have same sign.

II) product of two regression coefficients = (Correlation Coefficient)² i.e. $b_{yx} \cdot b_{xy} = r_{xy}^2$

III) The two regression lines intersect at the point $(\bar{x}, \bar{y})$.

IV) The Gradient of 'regression line of y on x ' is b_{yx} and the Gradient of 'regression line of x on y ' is $1/b_{xy}$.

Q From the following Data Obtain two Regression equations,

Sales	91	97	108	121	67	124	51	73	111	57
Purchases	71	75	69	97	70	91	39	61	80	47

Hence Estimate the Purchase when Sale will be 100. To get a purchase of 60 what is the required sales?

Ans

x	y	$u = (x - 90)$	$v = (y - 70)$	u^2	v^2	uv
91	71	1	1	1	1	1
97	75	7	5	49	25	35
108	69	18	-1	324	1	-18
121	97	31	27	961	729	837
67	70	-23	0	529	0	0
127	91	34	21	1156	441	714
51	39	-39	-31	1521	961	-1209
73	61	-17	-9	289	81	-153
111	80	21	10	441	100	210
57	47	-33	-23	1089	529	-755
		$\sum u_i = 0$	$\sum v_i = 0$	6360	2868	3900

$$\bar{u} = \frac{1}{n} \sum u = \frac{1}{10} \times 0 = 0$$

$$\bar{v} = \frac{1}{n} \sum v = \frac{1}{10} \times 0 = 0$$

$$\sigma_u^2 = \text{var}(u) = \frac{1}{n} \sum u_i^2 - \left(\frac{1}{n} \sum u_i \right)^2 = \frac{1}{10} \times 6360 - 0$$

$$= 636$$

$$\sigma_v^2 = \text{var}(v) = \frac{1}{10} \times 2868 - 0$$

$$= 286.8$$

$$\text{Cov}(u, v) = \frac{1}{n} \sum u_i v_i - \bar{u} \bar{v}$$

$$= \frac{1}{10} \times 3900 - 0 \times 0 = 390$$

$$\tau_{uv} = \frac{\text{Cov}(u, v)}{\sigma_u \sigma_v} = \frac{390}{\sqrt{636} \times \sqrt{286.8}} = 0.913$$

$$u = x - 90$$

$$\Rightarrow \bar{u} = \bar{x} - 90$$

$$\Rightarrow 0 = \bar{x} - 90$$

$$\Rightarrow \bar{x} = 90$$

$$v = y - 70$$

$$\Rightarrow \bar{v} = \bar{y} - 70$$

$$\Rightarrow \bar{y} = 70$$

we know,

$$y = \frac{x - c}{d} \Rightarrow \sigma_y = \frac{\sigma_x}{d}$$

Now,

$$v = \frac{x - 90}{1}$$

$$\Rightarrow \sigma_v = \frac{\sigma_x}{1} \Rightarrow \sigma_v = \sigma_x$$

$$u = x - 90$$

$$\Rightarrow \sigma_u = \sigma_x$$

$$v = y - 70$$

$$\Rightarrow \sigma_v = \sigma_y$$

$$\therefore \sigma_x = \sqrt{636}$$

$$\therefore \sigma_y = \sqrt{286.8}$$

$$u = x + b$$

$$v = cy + d$$

$$\tau_{uv} = (\pm) \tau_{xy}$$

$$a = 1 \text{ \& } c = 1$$

So,

$$\tau_{uv} = + \tau_{xy} = 0.913$$

we have,

$$u = +x - 90 \quad a = 1$$

$$v = +y - 70 \quad c = 1$$

∴ The Regression Coefficient of y on x ,

$$\begin{aligned} &= b_{yx} = r_{xy} \frac{\sigma_y}{\sigma_x} \\ &= 0.931 \times \frac{\sqrt{286.8}}{\sqrt{636}} \\ &= 0.61 \end{aligned}$$

Similarly,

The Regression Coefficient of x on y ,

$$\begin{aligned} &= b_{xy} = r_{xy} \frac{\sigma_x}{\sigma_y} \\ &= 0.931 \times \frac{\sqrt{636}}{\sqrt{286.8}} \\ &= 1.36 \end{aligned}$$

So, The Regression Line of y on x
 $\Rightarrow y - 70 = 0.61(x - 90)$

The Regression Line of x on y
 $\Rightarrow x - 90 = 1.36(y - 70)$

Now,

$$x = 100 \text{ (Given)}$$

$$\begin{aligned} y - 70 &= 0.61(100 - 90) \\ &= 6.1 \end{aligned}$$

$$\Rightarrow y = 76.1$$

$$y = 60 \text{ (Given)}$$

$$\begin{aligned} x - 90 &= 1.36(60 - 70) \\ \Rightarrow x &= (90 - 13.6) = 76.4 \end{aligned}$$

- (*) For two variables X and Y the equations of two Regression lines are $(x+4y+3)=0$ and $(4x+9y+5)=0$. Identify which one is of y on x . Find the means of x and y . Find the Correlation Coefficient between x and y . Estimate the value of x when $y = 1.5$.

Ans

Let us Assume that

$x+4y+3=0$ be the Regression Line of x on y
 & $4x+9y+5=0$ be the Regression Line of y on x

Now,

$$4x+9y+5=0$$

$$\Rightarrow 9y = -4x-5$$

$$\Rightarrow y = -\frac{4}{9}x - \frac{5}{9}$$

$$\text{slope} = -\frac{4}{9} = b_{yx}$$

$$x+4y+3=0$$

$$\Rightarrow 4y = -x-3$$

$$\Rightarrow y = -\frac{1}{4}x - \frac{3}{4}$$

$$\text{slope} = -\frac{1}{4} = \frac{1}{b_{xy}}$$

$$\Rightarrow b_{xy} = -4$$

Now,

$$b_{xy} \cdot b_{yx} = r_{xy}^2$$

$$\Rightarrow (-4) \times (-\frac{4}{9}) = r_{xy}^2$$

$$\Rightarrow \frac{16}{9} = r_{xy}^2$$

$$\Rightarrow r_{xy} = \pm \left(\frac{4}{3}\right) = -\frac{4}{3} \quad \left[\text{As } b_{yx} \text{ \& } b_{xy} \text{ has negative sign} \right]$$

But, $r_{xy} = -\frac{4}{3}$ is not possible because,

$$-1 \leq r_{xy} \leq 1.$$

Therefore Our Assumption was wrong.

So, $x + 4y + 3 = 0$ is regression line of y on x
& $4x + 9y + 5 = 0$ is regression line of x on y

Since, the point of intersection of two equations,
is $\bar{x}$ and $\bar{y}$, by solving two equations,

$$\bar{x} = 1$$

$$\bar{y} = -1$$

Regression Line of y on x slope $= -\frac{1}{4} = b_{yx}$

Regression Line of x on y slope $= -\frac{4}{9} = \frac{1}{b_{xy}}$
 $= b_{xy} = -\frac{9}{4}$

$$\sigma_{xy}^2 = -\frac{1}{4} \times \left(-\frac{9}{4}\right)$$
$$= \frac{9}{16}$$

$$\Rightarrow \sigma_{xy} = \sqrt{\frac{9}{16}} = \pm \left(\frac{3}{4}\right)$$

$$= -\frac{3}{4} \quad [\text{All other signs are Also Negative}]$$

$$y = 1.5 \text{ (Given)}$$

$$4x + 9y + 5 = 0$$

$$\Rightarrow 4x + (9 \times 1.5) + 5 = 0$$

$$\Rightarrow 4x = -13.5 - 5$$
$$= -18.5$$

$$\Rightarrow x = -4.625$$

Curve Fitting

④ Fit a straight line $y = a + bx$ to the following data :

x	2	5	6	8	9
y	8	14	19	20	31

Predict y when, $x = 9.6$.

Ans

x_i	y_i	$x_i y_i$	x_i^2
2	8	16	4
5	14	70	25
6	19	114	36
8	20	160	64
9	31	279	81
30	92	639	210

$$n = 5$$

∴ The Normal equations are,

$$92 = 5a + 210b \quad \text{--- (I)}$$

$$639 = 30a + 210b \quad \text{--- (II)}$$

$$\begin{aligned} \sum y_i &= na + b \sum x_i \\ \sum x_i y_i &= a \sum x_i + b \sum x_i^2 \end{aligned}$$

by solving equation (I) and (II)

$$a = 1, \quad b = 2.9$$

Now, The equation of best fitted straight line of the form $y = a + bx$ is,

$$y = 1 + 2.9x$$

$$\Rightarrow y = 1 + 2.9(9.6) \quad [x = 9.6 \text{ Given}]$$
$$= 28.84$$

Q. Fit a Second Degree parabola, $y = a + bx + cx^2$, to the following Data,

x	0	1	2	3	4
y	1	5	10	22	38

Predict y when, $x = 4.8$

Ans

$$\begin{aligned} a n + b \sum x_i + c \sum x_i^2 &= \sum y_i \\ a \sum x_i + b \sum x_i^2 + c \sum x_i^3 &= \sum x_i y_i \\ a \sum x_i^2 + b \sum x_i^3 + c \sum x_i^4 &= \sum x_i^2 y_i \end{aligned}$$

x_i	y_i	x_i^2	x_i^3	x_i^4	$x_i y_i$	$x_i^2 y_i$
0	1	0	0	0	0	0
1	5	1	1	1	5	5
2	10	4	8	16	20	40
3	22	9	27	81	66	198
4	38	16	64	256	152	608
10	76	30	100	354	243	851

The Normal equations are,

$$5a + 10b + 30c = 76 \quad \text{--- (I)}$$

$$10a + 30b + 100c = 243 \quad \text{--- (II)}$$

$$30a + 100b + 354c = 851 \quad \text{--- (III)}$$

By Solving (I), (II) & (III)

$$a = \frac{10}{7}, \quad b = \frac{17}{70}, \quad c = \frac{31}{14}$$

Now, the Best fitted parabola,

$$y = \frac{10}{7} + \frac{17}{70}x + \frac{31}{14}x^2$$

$$\Rightarrow y = 53.61 \quad (x = 4.8 \text{ Given})$$

* Fit an exponential Curve of form $y = ab^x$ to the following data.

x	2	3	4	5	6	7
y	640	512	410	328	262	210

Ans.

x_i	y_i	$Y_i = \log_{10} y_i$	x_i^2	$x_i Y_i$
2	640	2.8062	4	5.6124
3	512	2.7093	9	8.1279
4	410	2.6128	16	10.4512
5	328	2.5159	25	12.5795
6	262	2.4183	36	14.5098
7	210	2.3222	49	16.2554
27		15.3847	139	67.5362

$$\Sigma y = An + B \Sigma x_i$$

$$\Sigma y_i x_i = A \Sigma x_i + B \Sigma x_i^2$$

Normal equations are,

$$15.3847 = 6A + 27B \quad \text{--- (I)}$$

$$67.5362 = 27A + 139B \quad \text{--- (II)}$$

By solving Equation (I) and (II),

$$A = 3.0$$

$$B = -0.0968$$

$$A = \log_{10} a = 3 \Rightarrow a = 1000$$

$$B = \log_{10} b = -0.0968 \Rightarrow b = 0.8002$$

So, the Best fitting Curve will be,

$$y = 1000 \times (0.8002)^x$$

⑧ Fit a Geometric Curve of form $y = ax^b$ to the following Data.

x	1	2	3	4	5
y	5.0	6.3	7.2	7.9	8.5

Ans

x_i	y_i	$Y_i = \log_{10} y_i$	$X_i = \log_{10} x_i$	X_i^2	$X_i Y_i$
1	5	0.649	0	0	0
2	6.3	0.799	0.301	0.091	0.240
3	7.2	0.857	0.477	0.228	0.409
4	7.9	0.898	0.602	0.362	0.540
5	8.5	0.930	0.699	0.489	0.650
		4.183	2.079	1.17	1.839

Here, $n=5$,

$$\begin{aligned} \sum Y_i &= An + b \sum X_i \\ \sum X_i Y_i &= A \sum X_i + b \sum X_i^2 \end{aligned}$$

Putting the values in Normal equation,

$$A = 0.7009$$

$$b = 0.3263$$

$$A = \log_{10} a = 0.7009$$

$$\Rightarrow a = 5.022$$

So, the Best Fitting Geometric Curve will be,

$$y = 5.022 x^{0.3263}$$

TEST OF SIGNIFICANCE

Test of Significance

Large sample Test
($n > 30$) (Z Test)

- Single Mean
- Difference of Means
- Single proportion
- Difference of proportion
- Single Standard Deviation (Chi square Test)
- Difference of Standard Deviation

Small Sample Test
($n \leq 30$) (t Test)

- Single Mean
- Difference of Mean
- Chi-square Test for Goodness of fit.

Z Test

one tailed test
/ one sided alternative hypothesis ($>$ or $<$)

1% Level of Significance
2.33

5% Level of Significance
1.64

Two Tailed Test/
Both sided alternative hypothesis ($\neq$)

1% Level of Significance
2.576

5% Level of Significance
1.96

Large Sample Test for Single Mean

- ⑧ A machine part was Designed to withstand an average pressure of 120 units. A random sample size of 100 from a large batch was tested and it's found that the average pressure which these parts can withstand is 105 units with a S.D. of 20 units. Test 5% level whether the batch meet the specification. Suppose the population has a Normal Distribution.

Ans. Population Mean $= \mu = 120$

Sample Size $= n = 100$ [Large Sample]

Sample Mean $= \bar{x} = 105$

Sample S.D $= 20$

For Large Sample,

Population S.D (σ) $\cong$ Sample S.D (s)

Null Hypothesis:

The batch meets the Specification

$$(H_0 : \mu = 120)$$

Alternative Hypothesis:

The batch Doesn't Meet the Specification

$$(H_1 : \mu < 120)$$

$$Z_{crit} = 1.64$$

$$\begin{aligned} \text{Test Statistics} &= \left| \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \right| \\ (Z_{cal}) &= \left| \frac{105 - 120}{20/\sqrt{100}} \right| \\ &= 7.5 \end{aligned}$$

Here, $Z_{cal} = 7.5 > Z_{crit} = 1.64$

H_0 is Rejected and H_1 is accepted

So, the Batch Doesn't meet the Specification.

Large Sample Test for Difference of Mean

LED Bulbs manufactured by a Company A and Company B gave the following results:

	Company A	Company B
No of Bulb used	100	100
Avg Life	1248	1300
S.D in Hour	93	82

Now, find whether the average Life of ^{two} Bulbs made by the two Companies are same. Test at 0.05. [$Z = 1.96$ & $Z = \infty$ is 0.025]

Sol

for Company A,

$$n_1 = 100$$

$$\bar{x}_1 = 1248$$

$$s_1 = 93$$

$$s_1 \cong \sigma_1$$

for Company B,

$$n_2 = 100$$

$$\bar{x}_2 = 1300$$

$$s_2 = 82$$

$$s_2 \cong \sigma_2$$

Level of Significance = 5 %

Null Hypothesis :

The Average life of two bulbs made by two Companies are the same

$$(H_0 : \mu_1 = \mu_2)$$

Alternative Hypothesis :

The Average life of two bulbs made by two Companies are not same.

$$(H_1 : \mu_1 \neq \mu_2)$$

$$Z_{crit} = 1.96$$

$$\text{Test Statistics} = \left| \frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right|$$

(Z_{cal})

$$= \left| \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \right| \quad (\mu_1 - \mu_2) = 0$$

$$= \left| \frac{52}{\sqrt{\frac{93^2 + 82^2}{100}}} \right|$$

$$= 4.19396 \approx 4.19$$

Here, $Z_{cal} = 4.19 > 1.96 = Z_{crit}$

H_0 is rejected and H_1 is accepted.

So, The Average life of two bulbs made by two Companies are not same.

Large Sample Test for Single Proportion

- ① A sample of size 600 persons selected at Random from a large City shows that percentage of male in samples is 53%. It is believed that male to total population ratio in the City is $1/2$. Test whether this belief is confirmed by observation.

$$[\text{Given, } \int_{-\infty}^{-1.96} \phi(z) dz = 0.025]$$

Ans $n = 600$

$$\alpha/2 = 0.025$$

$$\alpha = 0.05$$

$$= 5\%$$

Level of Significance = 5%.

Sample proportion of male = $\hat{p} = 0.53$

Population proportion of male = $p = 0.5$

Null Hypothesis :

The belief is

Confirmed

$$(H_0 : P = \frac{1}{2})$$

Alternative Hypothesis :

The belief is not Confirmed

$$(H_1 : P \neq \frac{1}{2}) \quad [2 \text{ tailed Test}]$$

$$Z_{crit} = 1.96$$

$$\text{Test Statistics } (Z_{cal}) = \left| \frac{\hat{P} - P}{\sqrt{\frac{P(1-P)}{n}}} \right| = 1.47$$

$$Z_{cal} = 1.47 < Z_{crit} = 1.96$$

So, H_0 is Accepted, H_1 is Rejected.

So the belief is Confirmed.

Large Sample Test for Difference of Proportion

- ④ In a Certain City 100 men in a Sample of 400 were found to be smokers. In another city the Number of smokers was 300 in a random sample of 800. Does this Indicate that there is Greater proportion of smokers in the second city than in first city? Test at 1% Level of significance.

Ans for city 1,

$$n_1 = 400$$

$$x_1 = 100$$

$$\hat{p}_1 = \frac{100}{400}$$

$$= \frac{1}{4}$$

for city 2,

$$n_2 = 800$$

$$x_2 = 300$$

$$\hat{p}_2 = \frac{300}{800} = \frac{3}{8}$$

$$p = \frac{x_1 + x_2}{n_1 + n_2} = \frac{100 + 300}{400 + 800} = \frac{1}{3}$$

Level of significance = 1%

Null Hypothesis :

Proportion of smokers in 2nd city is not Greater than the first city $H_0 : p_1 = p_2$

Alternative Hypothesis :

Proportion of smokers in 2nd city is Greater than the first city

$$H_1 : p_1 < p_2$$

[one tailed Test]

$$Z_{crit} = 2.33$$

$$\text{Test Statistics } Z_{cal} = \left| \frac{p_1 - p_2}{\sqrt{p(1-p) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \right|$$

$$= 9.1856$$

$$Z_{cal} = 9.18 > 2.33 = Z_{crit}$$

So, H_0 is rejected and H_1 is accepted.

So, proportion of smokers in the 2nd city is Greater than first city.

Large Sample Test for Single Standard Deviation

- ⑧ A Random Sample of size 31 from a Normal Population Gives a Sample mean 42 and a Sample S.D of 5. Test at 5% level the hypothesis that the population S.D is 7. [Given $\chi^2_{0.05} = 43.77$ with 30 degrees of freedom].

Ans Here,

$$n = 31$$

$$\bar{x} = 42$$

$$s = 5$$

$$\begin{aligned} \text{Degree of freedom} &= (n - 1) \\ &= (31 - 1) \\ &= 30 \end{aligned}$$

Null Hypothesis:

Population S.D May be 7
($H_0: \sigma = 7$)

Alternative Hypothesis:

Population S.D May not be 7
($H_1: \sigma \neq 7$)

$$\chi^2_{\text{crit}} = 43.77$$

$$\begin{aligned} \text{Test Statistics} &= \frac{\sum (x_i - \bar{x})^2}{\sigma^2} & S^2 &= \frac{1}{n} \sum (x_i - \bar{x})^2 \\ (\chi^2_{\text{cal}}) &= \frac{n s^2}{\sigma^2} & \Rightarrow \sum (x_i - \bar{x})^2 &= n s^2 \\ &= \frac{31 \times 5^2}{7^2} & &= 15.82 \end{aligned}$$

Hence,

$$\chi^2_{cal} = 15.82 < 43.77 = \chi^2_{crit}$$

So, H_0 is accepted, and H_1 is rejected.

So, population S.D may be 7.

Large Sample Test for Difference of Standard Deviation:

⑩ Intelligence test on two Groups of Boys and Girls gave the following results:

	Mean	S.D	Number
Boys	70	20	250
Girls	75	15	150

Is there a significant difference in the standard deviation of the scores obtained by boys and girls? Test 1% Level of significance.

Ans for Boys,

$$n_1 = 250$$

$$\bar{x}_1 = 70$$

$$S_1 = 20$$

for Girls,

$$n_2 = 150$$

$$\bar{x}_2 = 75$$

$$S_2 = 15$$

Level of significance $\rightarrow 1\%$

Null Hypothesis:

There is no significant difference in the S.D of scores of Boys and Girls.

$$(H_0 : \sigma_1 = \sigma_2)$$

Alternative Hypothesis :

There is significant Difference in S.D of the scores of Boys and Girls.

$$(H_1 : \sigma_1 \neq \sigma_2) \text{ [Two tailed Test]}$$

$$Z_{crit} = 2.576$$

$$\begin{aligned} \text{Test Statistics (Zcal)} &= \left| \frac{S_1 - S_2}{\sqrt{\frac{S_1^2}{2n_1} + \frac{S_2^2}{2n_2}}} \right| \\ &= \left| \frac{20 - 15}{\sqrt{\frac{400}{500} + \frac{225}{300}}} \right| = 4.016 \end{aligned}$$

Here,

$$Z_{cal} = 4.016 > 2.576 = Z_{crit}$$

So, H_0 is rejected and H_1 is accepted.

So, there is significant Difference in the Standard Deviation of scores of Boys and Girls.

Small Sample Test for Single Mean

There can be two cases:

Case I : Population standard Deviation
Not Given in Question (σ)
[We will use t Test]

Case II : Population standard Deviation
is Given in Question
[We will use z test]

Case (I) :

⑩ Certain pesticide is packed into Bags by a Machine. A Random sample of 10 Bags is Drawn and their contents are found in weight (in kg) as follow: 50, 49, 52, 44, 45, 48, 46, 45, 49, 45. Test if average packing can be taken to be 50 kg. [$t_{0.25, 9} = 2.26$]

Ans

$$n = 10$$

x_i	$y_i = \frac{x_i - c}{d} \quad [c=45]$	y_i^2
50	5	25
49	4	16
52	7	49
$\vdots$	$\vdots$	$\vdots$
45	0	0

$$\sum y_i = 23$$

$$\sum y_i^2 = 117$$

$$y_i = \frac{x_i - c}{d}$$

$$\Rightarrow \bar{y} = \frac{\bar{x} - c}{d}$$

$$d = 1, c = 45$$

$$\Rightarrow \bar{y} = \bar{x} - 45$$

$$\Rightarrow \bar{x} = \bar{y} + 45$$

$$S.D(y) = \frac{S.D(x)}{d}$$

$$\sum y_i = 23$$

$$\Rightarrow \bar{y} = \frac{1}{10} \times 23 = 2.3$$

$$\text{So, } \bar{x} = 2.3 + 45 = 47.5$$

$$\begin{aligned} S.D \text{ of } y &= \sqrt{\frac{1}{n} \sum y_i^2 - (\bar{y})^2} \\ &= \sqrt{\frac{1}{10} \times 117 - (2.3)^2} \\ &= 2.53 \end{aligned}$$

$$S.D \text{ of } y = \frac{S.D \text{ of } x}{d}$$

$$\Rightarrow 2.53 = \frac{S.D \text{ of } x}{1}$$

$$\Rightarrow S.D \text{ of } x = 2.53$$

Null Hypothesis:

Population Mean is 50

$$(H_0 : \mu = 50)$$

Alternative Hypothesis:

population mean can't be 50.

$$(H_1 : \mu \neq 50)$$

$$t_{crit} = 2.26$$

$$\begin{aligned} \text{Test Statistics } (t_{\text{cal}}) &= \left| \frac{\bar{x} - \mu}{s / \sqrt{n-1}} \right| \\ &= \left| \frac{47.5 - 50}{2.5 / \sqrt{10-1}} \right| \\ &= 3.20 \end{aligned}$$

$$t_{\text{cal}} = 3.20 > 2.26 = t_{\text{crit}}$$

H_0 is Rejected, and H_1 is Accepted.

So, the Average weight can't be 50.

Case - (II)

⑩ A tyre manufacturing Company claims that the average life of their product is 69k miles. A Car manufacturing Company planning to purchase tyre, ^{they} collected a sample of 10 tyre's life from the population. The life of tyres are, 65, 72, 64, 71, 70, 69, 64, 63, 67 and 68 thousand miles. It is known that the population is normal and standard Deviation of population is $\sqrt{7.056}$. Test at 5% Level of Significance, that the average life of tyre of the company is lower than they claim. [Given, $Z_{0.5} = 1.65$, $Z_{0.1} = 2.36$]

Ans

$$n = 10$$

$$\sigma = \sqrt{7.056}$$

$$\bar{x} = \frac{1}{n} \sum x_i$$

$$= \frac{1}{10} [65 + 71 + 65 + \dots + 68]$$

$$= 67.2$$

Level of Significance = 5 %

Null Hypothesis :

The average life of tyre is not lower than 69.

$$(H_0 : \mu \geq 69)$$

Alternative Hypothesis :

The average life of tyre is lower than 69.

$$(H_1 : \mu < 69) \quad [\text{one tailed Test}]$$

$$Z_{\text{crit}} = 1.65$$

$$\text{Test Statistics} = \left| \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} \right|$$

(Z_{cal})

$$= 2.14$$

$$Z_{\text{cal}} = 2.14 > 1.65 = Z_{\text{crit}}$$

H₀ rejected and H₁ accepted.

So, Avg life of tyre is lower than 69.